

Blackout

AP Physics 2 · circuits & induction

Calder Switching Station, and two circuits of lattice towers walking off the edge of the map.

45	11	6.6	9/11	2	16·27·2
STOPS	CHOICE (24%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT·CONSTRUCT·OPERATE

READING Where this campaign stands

In good shape: 9 of 11 computed, the third-best mix in the set, and only two select-only concepts. One gap sits on a CONTROL and may be a Phase 0 phantom; the voltage-drop equation on stops 7 and 8 probably is not. The two select-only concepts are the two that matter most for a switching station — earthing and step-and-touch potential, and cascading failure — and both are currently one stop of multiple choice. **Read [plans/blackout-sequence.html](#) first.** Format mix and equations owed are this document's question; **when** the course teaches each concept against where the story is standing is a different one, and it is the worse of the two here — day 1 asks about droop control and the concept droop is built out of is claimed by no stop until day 12. Twelve of those inversions rest on material this campaign itself teaches later, and the fix is not a re-order.

NOW The mix as it ships

BALLPARK 11	CHOICE 11	PROTOCOL 6	CASEBOOK 3	SEQUENCE 2	VALUE 2	SCIENCETANK 2	DELEGATE 1	TRACE 1	TRIGGER 1	CONTROL 1	DEGENERACY 1	CHAIN 1	SWEEP 1	ATTEST 1
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CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$N_1/N_2 = V_1/V_2$	the transformer ratio, and why transmission is at high voltage	2·CHOICE 23·CONTROL 26·PROTOCOL	verify first
$\Delta V \approx I(R \cos \phi + X \sin \phi)$	voltage drop along a line, and why reactive power moves it	14·PROTOCOL 21·BALLPARK	real gap

1 of 2 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 1 is unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM	CONCEPT	STOPS
25	Earthing,	step-and-touch potential and safety	37
28	Cascading failure and the sequence of events		34

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects. The syllabus also lists 1 concept no stop touches at all — **Real, reactive and apparent power** — which is expected, and is a different conversation from this one.

FUN What this subject and this place can carry

HOLD STRONG

OPS · System Operations

Hold frequency inside a closing band while load steps arrive. The game **opens** on the swing equation; HOLD is that equation with the player's hand on it, and grid control is the canonical instance of the format anywhere in the catalogue.

SERVES the swing equation as a rate rather than a formula
NEEDS disturbances integrated with the control untouched; a do-nothing run must fail

TRIAL STRONG

the switchyard

A switching order driven through the yard, graded on order and not the clock: isolate, prove dead, earth, then work. Getting that order wrong is the accident the industry is built around, and concept 25 is select-only.

SERVES concept 25 — earthing, step-and-touch potential and safety
RISK world work; the yard and gatehouse already exist

PROBE STRONG

stop 6 or 7 · OPS / DIST

Take readings feeder by feeder along a line and find where the voltage collapses — which is not the point with the lowest number. Concept 28, cascading failure, is select-only, and a cascade is a chain you walk.

SERVES concept 28 and the voltage-drop equation
NEEDS one honest break and a decoy

SPOT STRONG

a TRANS or DIST stop

A switching instruction replaced mid-shift. Control-room work runs on standing instructions and the cost of missing a revision is measured in seconds, which is what the format weights.

SERVES operating discipline

TALLY MEDIUM

LOAD · Forecasting

Demand samples accumulate into a forecast that firms up. Right subject; competes with the game's existing SWEEP and its ten BALLPARKs.

BELT DEFER

A spares-and-cable store is a sort, and it teaches inventory rather than electricity. Defer.

LOB REJECT

Nothing is aimed. Reject.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: curriculumDelivery.mjs blackout --snapshot before.json , then --against before.json .

Minimum honest 3 STOPS · ONE SITTING

37	d13	DIST	CHOICE → SWEEP	The ground is not at one voltage the ground is not at one voltage — plot it and see
34	d12	OPS	CHOICE → PROBE	The failure that has not happened the contingency found by walking the feeders rather than read off a screen
2	d1	GEN	CHOICE → CONTROL	The reason the wires are not at wall voltage why the wires are not at wall voltage — change one thing, reverse it

CHOICE 11 → 8 (18%) · moves 16·27·2 → 13·27·5 · effective formats 6.6 → 7.9 · 3 new operate stops

Balanced 6 STOPS · THE RECOMMENDATION

37	d13	DIST	CHOICE → SWEEP	The ground is not at one voltage
34	d12	OPS	CHOICE → PROBE	The failure that has not happened
2	d1	GEN	CHOICE → CONTROL	The reason the wires are not at wall voltage
21	d7	DIST	BALLPARK → HOLD	Why the far end sits low the far end held up while load steps arrive
26	d9	METER	PROTOCOL → ATTEST	What is stored, and for how long what is stored and for how long — the record is not the condition
14	d5	TRANS	PROTOCOL → CHAIN	What actually limits a conductor what actually limits a conductor

CHOICE 11 → 8 (18%) · moves 16·27·2 → 13·26·6 · effective formats 6.6 → 9.1 · 4 new operate stops, 1 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

37	d13	DIST	CHOICE → SWEEP	The ground is not at one voltage
34	d12	OPS	CHOICE → PROBE	The failure that has not happened
2	d1	GEN	CHOICE → CONTROL	The reason the wires are not at wall voltage
21	d7	DIST	BALLPARK → HOLD	Why the far end sits low
26	d9	METER	PROTOCOL → ATTEST	What is stored, and for how long
14	d5	TRANS	PROTOCOL → CHAIN	What actually limits a conductor
23	d8	METER	CONTROL → TRIAL	A reading taken without a connection the switching order, driven — isolate, prove dead, earth
30	d10	DIST	CASEBOOK → SPOT	Why it cannot go faster an instruction revised mid-shift
40	d14	METER	CHOICE → TALLY	Hours over limit, added up demand accumulated until the forecast can be signed

CHOICE 11 → 7 (16%) · moves 16·27·2 → 11·26·8 · effective formats 6.6 → 10.0 · 7 new operate stops, 3 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. 2. Snapshot. 3. Edit the book, never the generated content — tools/import-book.mjs books/blackout.yml blackout --verify. 4. npm run traps: every instrument's trap must fire. 5. npm run check blackout and npm run drive blackout, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. 6. --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from `themes/blackout/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=blackout npm run dev`.